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REVIEW

Genetic alterations and clinical dimensions of oral cancer: a review

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Abstract

Oral cancer ranks sixth most prevalent type of cancer worldwide due to its alarming increase every year. Progression of oral cancer depends on various heterogeneity pathways, but their exact mechanism remains unclear. Genetic aberrations on oral cancer cells set back the effectiveness of existing therapies and make it more challenging by triggering drug resistance. To understand the intricate details of oral cancer pathogenesis and for advancing current therapies, genetic modifications are the most promising approach. In this review, we tabulated the information on genetic alterations, microbial associations, aberrant signalling pathways and their clinicopathological characters on the pathogenesis of oral cancer. We primarily discussed the pitfalls of the current treatment regimen and its associated drug resistance pattern, which will provide a clear insight into developing new drugs. We also highlighted the genetic-molecular targets with their current clinical status on drug development and its outcome.

Keywords Oral squamous cell carcinoma · Microbes · Genes · Therapeutics · Drug resistance · Clinical phases

Introduction

Head and neck cancer, the sixth prevalent cancer worldwide begins in the epithelial lining of the upper-digestive tracts such as the oral cavity and larynx [1]. Based on its position and cells that get affected, it is graded as oral and pharyngeal cancer, squamous cell carcinoma and adenocarcinoma. Cancer patterns vary across different regions of the world where oral squamous cell carcinoma (OSCC) or oral cancer ranks the first in infecting males, third among females especially in developing countries like India, Sri Lanka, and Bangladesh owing to their lack of knowledge about the consequences of tobacco and alcohol [1]. Recently, developed countries also reported increased OSCC cases due to changes in their lifestyle. The second most prevalent cancer in transplant patients is OSCC (e.g., Leukaemia, multiple myeloma, and lymphoma). It is a curable disease when noticed early, but inevitably mortal when diagnosed too late. Surgery in the initial stages and combination therapy (chemotherapy, radiotherapy) in later stages are the selected treatments for oral cancer [2]. Despite these available treatments, survival rate is not improved significantly as the accumulation of genetic mutations impaired the effectiveness of drugs through therapeutic resistance.

Oral cancer is a heterogeneous disease common in males than females caused due to the mutation in tumour suppressor genes, oncogenes, DNA repair genes, immune-inflammatory genes and other genes, in association with the exposure of tobacco, betelnut, virus and bacteria [3]. Among these, alteration in EGFR, cyclin D1, H-Ras and AKT led to the development of EGFR inhibitors, CDK inhibitors and mTOR inhibitors as new therapeutic strategies for OSCC. Despite the significant progress in understanding epidemiology, pathology, molecular processes and therapeutic strategies, the burden remains high with an increase in mortality due to the lack of understanding in complex molecular pathways and different modes of drug resistance involved in treatments. In this review, we primarily discussed the external and biological factors concerning OSCC molecular pathways. Then, we investigated the complex genomic aberrations that drive OSCC as the details of molecular pathways are uncovered if their translational importance is unknown. We also discussed the standard treatments and role of tumour suppressor genes, oncogenes, DNA repair genes in providing intrinsic and acquired drug resistance.

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We aim to highlight progress in molecular and clinical research, reflecting the current understanding of OSCC and their therapeutic strategies that may contribute to the discovery of new therapeutics.

External risk factors for OSCC

Tobacco as a risk factor

More than 1,80,000 incidences of oral cancer arise in South East Asia each year, of which 90% are a result of smoking and chewing habits. There are early demonstrations from the year 1964 on tumour causing property of tobacco, but molecular events in tumour development due to its ingestion are not well analyzed. Tobacco can contain more than 60 proven or potential carcinogens thereby increasing cancer risk over different mechanisms [4]. The compounds 4-(nit rosomethylamino)-1-(3-pyridyl)-1-butanone (NNK) and N′-nitrosonornicotine (NNN) present in tobacco causes millions of death annually [5]. Early reports suggest that mutation in carcinogen metabolizing enzymes like Cytochrome p450 and glutathione-S-transferase enzyme can be a prominent cause of oral cancer [6]. Tobacco also causes methylation in DNA repair genes like *p53, p16, O^6-methylguanine DNA methyltransferase* (*MGMT and human mut L homologue 1* (*hMLH1*) [7]. Besides, Betel nut is responsible for complications like oral submucous fibrosis, other than OSCC, especially in Asian people [8]. Alcohol, independent risk factor remains controversial, but it is proven that it aids carcinogens to penetrate oral mucosa. The presence of carcinogens like mycotoxins, urethane and accumulation of acetaldehyde is associated with alcohol caused oral cancer [9]. OSCC is generally preceded by oral potentially malignant disorders like erythroplakia, leukoplakia and oral submucous fibrosis as a result of reactive oxygen species (ROS) and associated metabolites released from tobacco, betel nut and alcohol. Few reports proved protein kinase B or AKT activation with diverse cellular functions plays a pivotal role in tobacco and alcohol-induced oral carcinogenesis [10]. However, there is a chance that an oral squamous surface will face several challenges at genetic and molecular levels as our mouth can have numerous microbes [11].

Microbes as a risk factor

Oral microbes increase the risk of oral cancer through the activation of alcohol and smoking- related carcinogens and are also involved in metastasis. It is stated that microbial infection in oral microbiota releases pro-inflammatory mediators facilitating oncogene activation, cell proliferation, apoptosis inhibition and direct release of carcinogens. A series of cultivation studies and sequencing studies on various microbes related to OSCC, including *Fusobacterium, Streptococcus, Parvimonas, Porphyromonas, Rothia, and Capnocytophaga* species were conducted between 1993 and 2018, but their results were not promising except *Porphyromonas gingivalis* [12]. In an in-vitro study, *Porphyromonas gingivalis* promoted immune evasion was confirmed through overexpressed programmed death-ligand 1 (PD-L1) and disrupted apoptosis through AKT activation [13]. Candidalysin, a toxin encoded by endothelin converting enzyme *1* (*ECE1*) gene of *Candida albicans,* promoted tissue invasion through activating cell proliferation pathways like mitogen-activated protein kinase (MAPK) and phosphoinositide 3-kinases (PI3K) causing epithelial damage and malignant lesions [14].

Multidisciplinary methods are also required to obtain a detailed understanding of the potential aspects of oral bacteria in tumour development and progression. OSCC is considered as the 2nd most common cancer caused by human papillomavirus 16 (HPV 16) but with better survival rates as it causes less genetic damage affecting p53 and pRb. In oral cavity, almost 36 types of HPV are associated with benign and malignant lesions, among which HPV16, Human herpes virus and Herpes simplex virus are said to be a foremost environmental factor in non-smokers [15, 16]. In comparison to the healthy oral epithelium, Epstein Bar virus (EBV) and Hepatitis C virus (HCV) had reported being identified recurrently in oral lesions such as lichen planus and OSCC. However, its pathogenicity remains elusive [17]. Many biological factors are found to have possible role in OSCC but the association of co-factors like tobacco is strongly required for oral carcinogenesis.

Genetics as a risk factor for OSCC

Heredity as a risk factor

Familial and subsequently almost certain hereditary characters have a significant role in cancer risk and may conjoin with ecological exposures. Oral malignant growth study led by two groups of researchers at different timespan revealed the presence of autosomal dominant inheritance. There is an increased risk for OSCC with a heritable condition like Fanconi anemia and bloom syndrome [18]. Recently, 3 kids in a family with no history of tobacco use were reported dead, only by mutations in cell adherence and angiogenesis regulatory genes *IQGAP1* and *VAV2* [19]. Though not many investigations have done so far, the connection between familial history and oral malignancy will form a breakthrough in the diagnosis.

Genetic Instability as a risk factor

Concerning solid tumours like OSCC, several functional and chromosomal abnormalities are explained in molecular pathways such as cell cycle control, signal transduction, apoptosis and angiogenesis. Chromosomal segregation, aneuploidy, loss of heterozygosity (LOH), telomere instability, mutations in cell-cycle checkpoints and DNA damage repair are well known genetic abnormalities involved in tumorigenesis. Loss of heterozygosity in Chr. 3p, 9p, 11q and 17p is consistent with prior reports and assumes a significant role [20].

Tumour suppressor gene

OSCC is a challenging pathology where mutation/amplification/methylation of tumour suppressor genes results in uncontrolled proliferation and invasion of cells. In both gene copies, tumour silencer genes had inactivated in a "two-hit" manner through point mutations, deletions, hypermethylation and rearrangements. On DNA damage ATM (ataxia-telangiectasia mutated) member of PI3K family enables p53 and regulates apoptosis. In a study, mutation at ATM rs189037 was found in OSCC samples suggesting their role in oral cancer [21]. Mutated p53 inhibits cell cycle inhibitors (p16. p52) and are the most frequently studied genetic errors in OSCC promoting cell proliferation. It connects with the clinical seriousness of the disease and prevalent in HPV positive tumour [22, 23]. A study conducted on Asian patients found p53 mutation less common in comparison with Caucasians [24].

P63 (3q) member of the p53 family, was also found to be highest with lack of differentiation along with poor prognosis [25]. Retinoblastoma (Rb), the first tumour suppressor gene identified is a key controller of G1 progression and possesses 16 potential sites of cyclin-dependent kinases (CDK) phosphorylation. Hyperphosphorylation or mutation of pRb leads to pRb overexpression thereby increasing Cyclin D1 expression resulting in OSCC development especially in smokers. Although several studies have been conducted to establish the role of Rb in OSCC, the exact mechanism behind the alteration and overexpression is left obscure [26].

Notch is the 4th highly interested protein involved in OSCC development. It plays a dual role as a tumour suppressor gene and oncogene based on the tissue involved and stage of cancer progression. Among all the notch receptors (Notch 1–4) Notch 1 and 4 are widely correlated with the clinical and T stage of OSCC. The components of the Notch signalling pathway (JAG 1 and 2, NCID, HES, HEY) that regulates cell differentiation are known to exist in the oral cavity and its functionality but not well studied [27, 28]. Fragile histidine triad diadenosine triphosphatase (FHIT,) which maintains cell homeostasis has found to be lost in the majority of smokers [29]. Downregulation of essential genes like cadherin catenin complex-1 (*CDH-1*), cyclin-dependent kinase 4 inhibitor C (*CDKNC*), CYLD lysine 63 deubiquitinase (*CYLD*), forkhead box P1 (*FOXP1*), phosphatase and tensin homolog (*PTEN*) and von Hippel-Linda (*VHL*) had been connected with epithelial to mesenchymal transition (EMT) and poor survival [30, 31]. *PTEN*, the negative regulator of the PI3K/AKT pathway is the most commonly deregulated genes resulting in the constitutive activation of cell survival signalling correlated with early stage of OSCC [32, 33]. Additionally, aberrant micro RNA like miR-223 and miR-92a expression promoted OSCC by downregulating FBXW7, FOXP [34]. On the contrary mutated cadherin 11 (*CDH11*), cyclin-dependent kinase-6 (*CDK6*), CCAAT enhancer-binding protein alpha (*CEBPA*), nucleophosmin1 (*NPM1*) are overexpressed. The clinicopathological features and genetic alterations of numerous tumour suppressor genes having a role in OSCC development are given in Table 1.

Proto-Oncogene

Numerous oncogenes are categorized in cancer, but relatively few are crucial in OSCC development and reportedly raise the number of oncogenes located in 1q, 3q, 5p, 7q, 8q, 9q, 11q, 12q, 14q and 15q regions. Mutation in signalling pathways like EGFR, FGFR, MET, RET, Notch, NFκB, and Wnt will allow their downstream signalling mechanisms to be constitutively triggered, resulting in cell proliferation and metastasis. Pathway assessment of 47 tumour samples revealed somatic aberrations within the OSCC genome involved in major pathways, including cell cycle and RTK-MAPK-PI3K [48]. Epidermal growth factor receptor (EGFR), a tyrosine kinase receptor expression, was 42 to 58% prominent among OSCC cases with dominant malignancy. Mutations in the EGFR gene is prevalent and activate several pathways like Ras/Raf/MEK/ERK, Ras/PI3K/PTEN/AKT/mTOR inducing transformation, proliferation, differentiation, migration and angiogenesis resulting in advanced oral tumours [49]. Research over the past two eras indicates *H-Ras* is the utmost genetically deregulated oncogenes than K-Ras and N-Ras in oral cancer. It activates several downstream targets regulating cell cycle and proliferation and is expressed diverse among various geographical regions [50]. C-myc transcription factor phosphorylated by ERK and Wnt pathway regulates human telomerase reverse transcriptase (hTERT), activates CDK promoting proliferation, inhibits cellular differentiation and interacts with other oncogenes to promote tumorigenesis with diminished Bcl-2 in later stages [51].

Phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha (*PI3KCA*) triggered by EGFR drives PI3K/AKT pathway, where overexpression, mutation and alteration of copy numbers result in constitutive activation of the

Table 1 Location and pathological characteristics of tumour suppressor genes in OSCC

Gene	Chromosome location	Alteration type	Clinical characteristics	Pathology characteristics	Reference
APC	5q21	Codon 1621 Exon 15	Smokers, Betelnut, HPV associated with p53 mutation	Lymph node metastasis (LNM)	[35]
ATM	11q22-23	SNP at rs19037, promotor hypermethylation	Smokers, betelnut	Poor prognosis	[21, 36]
CDH1	16q22.2	G/GA-347, promotor hypermethylation	Betelnut, tobacco	LNM	[37]
CDKN2C	1p32.3	c.238C > T p.Arg80Ter	Tobacco, HPV	Large tumour, recurrence, poor survival	[38, 39]
CYLD	16q	D168N	HPV, tobacco	Invasion and inhibit innate immunity, induce cisplatin resistance	[40, 41]
FBXW7	4q32	p.S462F	Tobacco	Contribute to radio resistance	[42, 43]
NPM1	5q35.1	P300 mediated acetylation	Tobacco, smoking, alcohol	TNM, advanced clinical stage	[44, 45]
PTEN	10q23	Exon 5,7 SNP at rs9651495	Smoking	Metastasis	[32, 39, 46]
TP53	17p	Arg72Pro, 2 intron and 11exon polymorphisms	Alcohol, tobacco, HPV	Extra node extension and advanced stages with low E-cadherin expression, cisplatin resistance	[47]

AKT pathway and confers radio resistance. Activated AKT enhances transcription of NFκB, regulates p53 through MDM2 (murine double minute), inactivates GSK-3β resulting in cyclin D1 overexpression and thereby promoting OSCC proliferation (Fig. 1) [52]. SHP2 encoded by *PTPN11* mediates AKT and ERK activation and is overexpressed. A study reported that increased expression of SHP2 amplified the metastasis regulator matrix metalloproteinase-2 (MMP-2) secretion, but the exact mechanism is unclear [53]. On the contrary, c-Jun N-terminal kinase (JNK) of MAPK's family that phosphorylate c-Jun was found to cause tumour and inhibition of JNK reduced cell proliferation and capillary tube formation [54]. C-met, the only tyrosine kinase receptor to bind hepatocyte growth factor (HGF), was confirmed to be hiking in cigarette users and was previously associated with increased survival and reduced metastasis in knockdown mice. In comparison, reports on 40 OSCC specimens confirmed that there was no correlation between cigarette and C-met modulation, but indicated that ERK/AKT/NFκB could be a potential C-met mediated mechanism [55] (Fig. 1). Also, in early clinical stages, fibroblast growth factor receptors 1 and 2 (FGFR1, FGFR2) are frequently amplified (17%) [56].

APC inactivation, EGFR phosphorylation or mutation in the Wnt signalling components causes aggregation of β-catenin in the nucleus resulting in activation of target genes such as *C-myc*, *c-Jun* and *CCND1*. Epigenetic silencing of Wnt-related genes by histone modification and DNA methylation of E-cadherin promoter, *CTNNB1* mutation, WNT antagonists like secreted frizzled-related protein 2 (SFRP-2), Wnt inhibitory factor 1 (WIF-1), dickkopf-related protein 1 (DKK-1), Dachshund homolog 1 (DACH1) and RUNX3, are significantly associated with tumour recurrence and poor prognosis [57]. NF-κB regulating immune and inflammatory genes contribute to invasion, survival, chemoresistance and angiogenesis are constitutively expressed. Reports suggest that NF-κB may contribute significantly to tumour development through increased expression of interleukins IL-6, IL-8 and matrix metalloproteinase-9 (MMP-9) [58]. The transcription factor CCND1 triggered by multiple oncogenic mutations in mitogen signalling pathways, Wnt pathway, NF-κB pathways, results in overexpression and excessive proliferation of cyclin D1. Cyclin D1 expression encoded by *CCND1* oscillate during the cell cycle plays a key role in the development of OSCC and PIK3-GSK3β stabilization. The most copy number alteration found in OSCC is *CCND1/PRAD1* amplification correlated with clinical stages [59].

It is documented that *int-2* gene homologous to FGFR is co-amplified with *hst-1* and contributes to oral tumorigenesis [60]. Human salivary peptide histatin-1 (hst-1) that acts as a protective layer for oral epithelium is altered in OSCC [61]. Adding to, it has stated that *RAF1, RET, REL, BCL-6, KIT, HMG* have a role in oral cancer [39, 62–65]. The clinicopathological features and genetic alterations of numerous oncogenes having a role in OSCC development are given in Table 2.

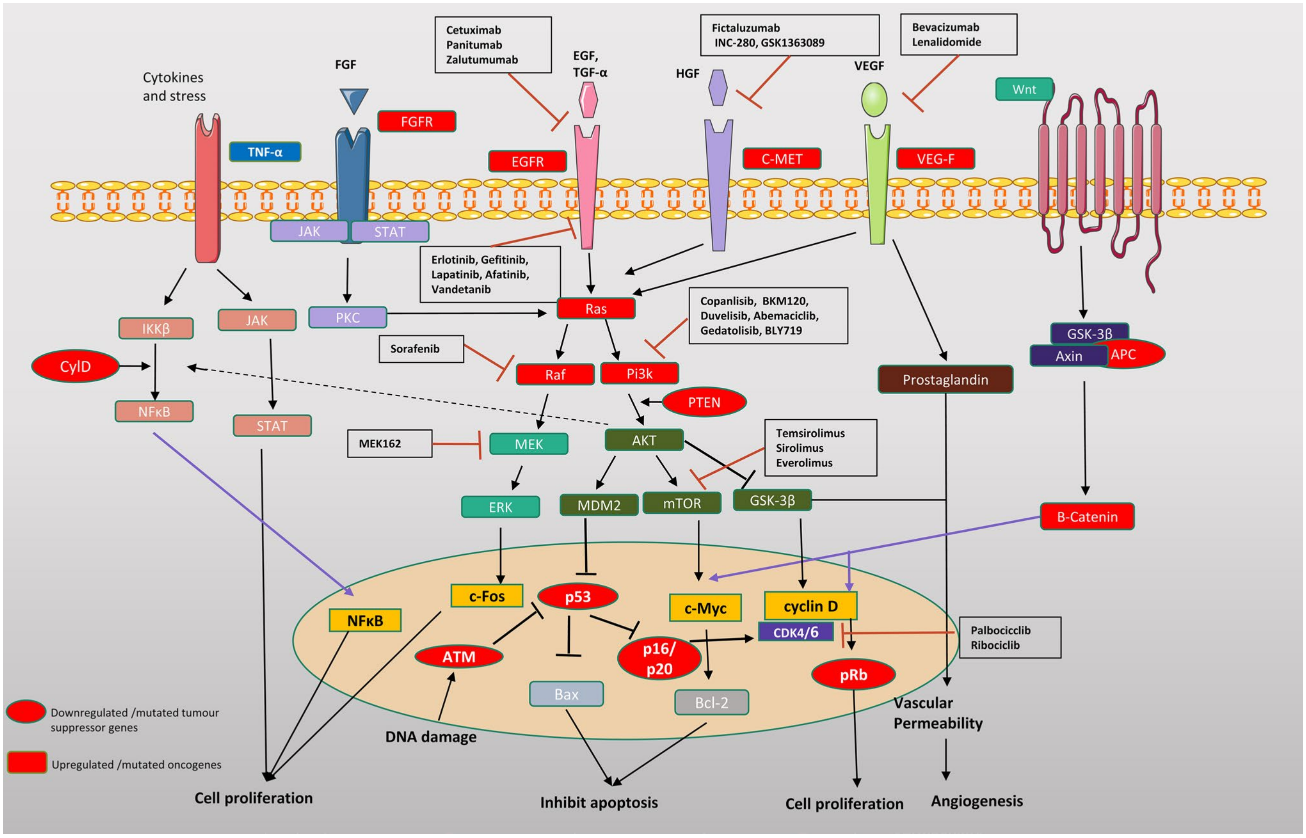

Fig. 1 Schematic diagram representation of various genes and associated pathways involved in OSCC and their clinical targets

DNA repair genes and immune genes

Many DNA damage repair pathways such as ataxia telangiectasia mutated (ATM)/ATR involve in p53 activation in response to DNA damage causing radioresistance. Mutation in these pathways inactivates p53, resulting in abnormal proliferation due to the process of replication in oral cancer [36]. Genes involved in Base Excision Repair (BER) are involved in the development of oral cancer, among which XRCC1 are extensively reported conferring radioresistance. There is a significant association between Arg194Trp, Arg399Gln polymorphism and OSCC development, specifically amid Asians [82]. *POLQ* encoding RNA polymerase is seen to upregulate in Brazil oral cancer samples [83].

Among the many signalling pathways, TGF-β signalling has a crucial role as it activates the cell proliferation controller called CDK inhibitors, amplifies the expression of MMP-1, MMP-3, MMP-9, MMP-10 and promote tumour invasion [84]. In 2019, a research group experimentally proposed that miR-103 promotes invasion and metastasis of OSCC cells (TCA8113) by upregulating MMP-2 and MMP-9 via SALL4 negative regulation [85]. The role of immunological cells such as tumour necrosis factor-related apoptosis-inducing ligand (TRAIL), dendritic cells, NK cells, macrophages, eosinophils and mast cells were studied in oral cancer to understand its function in tumour progression. A study documented that the loss of TRAIL expression is an initial occurrence during oral carcinogenesis and may be involved in apoptosis dysregulation and lead to OSCC molecular pathogenesis [86]. The abnormal expression of key inflammatory mediators Cyclooxygenase-2 (COX-2) activates MMP-7, causing invasion and metastasis. COX-2 is widely expressed in oral cancer and oral premalignant lesions and appears to be particularly enhanced in high-risk oral lesions [87]. PD-L-1 is one of the major genes overexpressed in OSCC, being linked to key signalling pathways such as NF-κB, ERK1/2, PIK, JAK/STAT and ROS in tumour tissue. Cluster of differentiation 274 (*CD274*) encoding PD-L1 has been noted for persuading death and recurrence in association with *P.gingivalis* [88]. Moreover, mast cells are shown to induce angiogenesis, proving the role of these immune cells in OSCC requires more comprehensive assessment.

Other genetic factors as a risk factor

Significant expression of Extracellular matrix metalloproteinase inducer (EMMPRIN), is proven to be important in cell proliferation and progression [89]. Telomeric

Table 2 Location and pathological characteristics of oncogenes in OSCC

Gene	Chromosome location	Alteration type	Clinical characteristics	Pathology characteristics	Reference
AKT-1	14q32	Exon 2 E17K	Tobacco, alcohol and HPV	Increased aggressiveness, migration	[46, 66]
BCL-2	18q21	ala43ala	Tobacco and betel quid	Poor prognosis, LNM, therapeutic resistance	[67]
CCND1	11q13	Numerical aberration at Exon 3, SNP at G870A, C/G1722	Tobacco, alcohol betel quid and HPV	Poor prognosis, aggressive tumour, LNM and invasive	[68, 69]
CTNNB1	3p22	Exon 3	Tobacco, areca quid and HPV	LNM, poor prognosis and Invasive	[70, 71]
DEK	6p22	–	Tobacco	LNM, poor survival	[72]
FGFR1	8p12	8p12 amplification	Men, tobacco, HPV	Early stage, LNM	[56]
FGFR4	5q35	rs351855 G/A G388R	Betel quid, tobacco	Stage III, IV and poor prognosis	[73]
H-Ras	11p15	Codon 12, 13, 61	Betel quid, tobacco and based on ethnicity	All stages, poor prognosis	[50, 74]
K-RAS	12p12	Codon 12	Betel nut	Stage IV	[50, 75]
Mdm2	12q15	SNP 309 T > G	HPV, tobacco, betel nut	Dysplastic lesions and poor prognosis	[47, 76]
Met	7q31	–	Tobacco	EMT, poor survival and poor prognosis	[55]
Notch 1	9q34.3	G481S E424K	Tobacco and alcohol	Oral lichen planus	[77]
C-myc	8q24	Amplification	Male, tobacco	Stage III, IV and poor prognosis	[51]
PIK3CA	3q26	Exon 9, 20 1633G > A E545K and other 15 mutations	Tobacco, betel quid and alcohol	Stage II and invasive, resistance to cetuximab and palbociclib	[78–80]
PTPN11	12q24	E76K	Tobacco	Poor prognosis, metastasis and invasion	[53, 81]

heterocomplex protein expression named shelterin appears in some of the oral cancer cases [90]. TC21, members of Ras family interact with Erk2 and PI3K has a crucial role in the early stages through initiating AKT, Bad, mTOR and glycogen synthase kinase 3 beta (GSK-3β) [91].

Regarding cytoskeletal defects, cytokeratin (CK1, CK5, CK13, CK16, and CK19) is found to downregulate, while CK8, CK14, and CK18 are supposedly upregulated [92]. Significant signalling pathways of tumorigenesis are regulated by several micro-RNAs as tumour suppressors or inducers. miR-31, miR-146a, miR-144, miR-7, miR-184, miR-127 on overexpression promotes OSCC. As a result of hypermethylation, miR-101, miR-26a/b, miR-29a, miR-27b, miR-137, miR-125a, miR-29a, miR491-5p, miR-124, miR-125, miR-218, miR-99a, miR-375 are downregulated [93]. Epigenetic alterations, including DNA methylation, histone modifications and non-coding RNA play a crucial role in OSCC growth. Recently long non-coding RNA named HOX transcript antisense RNA (HOTAIR) and Ferritin heavy chain 1 pseudogene 3 (FTH1P3) were said to cause metastasis and poor survival [94]. Besides *FOXCUT, MALAT1, UCA1, TUG1, CCAT2, FTH1P3, H19,* and *HIF-CAR / MIRHG* are downregulated.

The involvement of Vitamin D Receptor (VDR) expression in oral tissue also represents VDR's role in affecting the etiology of oral cancer and proves that vitamin D could inhibit proliferation and tumour cell differentiation [95]. Circadian genes that normally regulate autophagy are poorly expressed in OSCC. In a study *Per 1* gene regulated autophagy-mediated cell proliferation through AKT/mTOR pathway activation [96]. Renin-Angiotensin system, besides its role in blood pressure maintenance and homeostasis, it is also involved in oral cancer tumorigenesis. Angiotensin receptors I, II are overexpressed in oral mucosa and oral cancer stem cells (OCSC). In an in-vitro study, metabolite Ang1-7 is proven as a therapeutic target in head and neck squamous cell carcinoma (HNSCC) [97].

Current treatment regimens and its mechanism in OSCC

Surgery, radiotherapy and chemotherapy are the standard methods of medicine for oral malignancy. The primary treatment of oral cancer is surgery and is associated with disfigurement, functional impairment, oral candidiasis and mucositis [98]. Adjuvant chemoradiotherapy is an option given to locally advanced unresectable oral cancer patients to decrease radiation resistance and recurrence during severe medical circumstances. Along with these, other complications like dysgeusia, osteoradionecrosis, soft tissue necrosis, gradual loss of periodontal contact, trismus and xerostomia are caused by radiotherapy. In locally advanced head and neck squamous cell carcinoma, the combination of TPF (Cisplatin, 5-FU and docetaxel) and chemoradiation are used to reduce recurrence and the efficacy remains poor with hematologic toxicity [99]. Generally, Cisplatin crosslinks with purine bases, 5-flourouracil (5-FU) block thymidine synthesis and docetaxel inhibit mitotic spindle assembly during mitosis leading to cell death [100]. For metastatic cases, Cisplatin or Carboplatin in conjunction with 5-FU, Docetaxel and Paclitaxel are prescribed as first-line treatment, in which Paclitaxel target microtubules and induce apoptosis. Also, Cetuximab combined with platinum-based drugs and 5-FU improved progression-free survival with a reduction in toxicity and thus approved by the FDA. Patients who are sensitive to platinum-based drugs are treated with Paclitaxel and Cetuximab. Currently practiced food and drug administration (FDA) approved drugs are Cisplatin, Carboplatin combined with Methotrexate, Docetaxel, Cetuximab with radiotherapy, platinum-based chemotherapy, Bleomycin, Hydroxyurea, Nivolumab, and Pembrolizumab. The overall effectiveness of traditional cytotoxic chemotherapy for OSCC has hindered by the lack of selectivity, narrow therapeutic range, and the widespread emergence of mechanisms for drug resistance [101].

Mechanisms of resistance in OSCC

Several regimens are in clinical phases but EXTREME (Cisplatin or Carboplatin with 5FU and cetuximab) accompanied by cetuximab has been the standard first-line treatment since the last decade for OSCC. In this regimen, Cisplatin or Carboplatin is the platinum agent forming DNA adduct. Cetuximab is the EGFR inhibitor that downregulates EGFR signals and facilitates cell-cycle arrest and 5-FU is an antimetabolite inhibiting thymidine synthetase leading to apoptosis [102]. The efficacy of this regimen is hampered due to various drug resistance mechanisms. Typically, the mechanism of drug resistance varies between various types of cancer and the kind of agents used. Specific transporters associated with reduced drug transport and increased efflux such as P-gp, MRP1, and ABCG2 prevent the drug from meeting the intracellular targets and contribute to acquired drug resistance [103]. Other factors such as tumour suppressor genes, oncogenes, DNA damage and repair genes, long non-coding RNAs, cancer stem cells also lead to innate and acquired resistance. Dysregulation in the anti-apoptotic protein Bcl-2, pro-apoptotic protein Bax and p53 was reported to develop chemoresistance in OSCC [104]. Cisplatin is the most used chemotherapy drug, but several mechanisms of drug resistance hinder its efficacy. Excision repair complimentary group 1 (ERCC1), nucleotide excision repair gene is said to be aberrated and involved in cisplatin resistance [105].

A microarray analysis carried out on Cisplatin-resistant cell line TCA8113 expressed 63 differential genes, among which *CCND1* and *CCND3* are widely reported [106]. NFκB and its target genes such as xIAP, Bcl-2 and Cox-2 also resulted in cisplatin resistance. In an *in-vitro* study, cisplatin-resistant cells when combined with quercetin downregulated xIAP, thus promoting apoptosis through the Akt-IKKβ-NFκB-xIAP [107]. Though Cisplatin has been investigated for over 50 years, the molecular mechanism behind the drug resistance in OSCC remains indistinct.

In the case of 5-FU, the mechanism of resistance in OSCC remains unclear. As overexpressed EGFR has conferred radioresistance in OSCC, Cetuximab is employed to induce apoptosis through blocking PI3K and JAK/STAT. Adding Cetuximab to platinum-based drugs and 5FU increased the rate of survival, but the efficacy remains disappointing due to various complex mechanisms leading to intrinsic and acquired resistance. Major mechanisms involved in cetuximab resistance are KRAS, MET, PIK3CA and HER2/3. Upregulation of HER2/HER3 of EGFR family that activates PI3K/AKT has been elucidated as the prominent mechanism for cetuximab resistance and simultaneous blocking of HER2 and HER3 was proposed as an important strategy for resistance in a patient-derived xenograft model [108]. In a cell line study, loss of p53 was contributed to Cetuximab and radioresistance. Reconstituting p53 sensitized the cells to Cetuximab, thereby suggesting the role of p53 in drug resistance [109]. CD44 which binds to hyaluronan activates the downstream signalling pathways MAPK and PI3K to induce shelterin mediated invasion. During epithelial to mesenchymal transition (EMT), CD44 expression increases conferring resistance to Cetuximab [110]. In another study, Cetuximab resistant cells had shown increased MET (HGF binding receptor), which activates AKT signalling and treating those cells with MET and Cetuximab inhibited cell growth by inactivating ERK and AKT [111]. This study provides a base for synergistic usage of MET inhibitor and EGFR inhibitor in Cetuximab-resistant OSCC. Furthermore, Fictaluzumab (C-Met inhibitor) is synergistically studied in combination with Cetuximab in phase I of HNSCC. (NCT02277197). Accumulating studies also provided the role of miRNA in Cisplatin chemoresistance among which miR-218, miR-24, miR-654 are widely reported [112]. Oral Cancer Stem Cells (OCSC) existing within the oral tumour also contributed to tumour initiation, chemotherapeutic resistance and recurrence. CD-44, cell surface glycoprotein overexpressed in OCSC had reported playing a vital role by influencing Raf-MEK and PIK3-GSK3β-AKT pathways [113]. All these data suggest that there are many reductant mechanisms for cancer cells to confer resistance to therapy. Also, these studies provide evidence that combinatorial therapy may be an alternative to overcome drug resistance.

Upcoming clinical considerations

Due to the advanced discovery of various genetic and molecular biology pathways, pharmacological agents and therapeutic antibodies are developed, in vision to selectively target crucial signalling molecules. Such methods are approved already to be used on certain forms of solid tumours and many clinical developments ongoing in OSCC are discussed here. Table 3 summarizes the various clinical trials of OSCC (https://clinicaltrials.gov/).

Pan-CDK inhibitors

CDK4/6 has extensively reported overexpressing in OSCC can be used in cancer care as therapeutic targets [114]. Pan-CDK inhibitors and new specific CDK inhibitors are the two types where Pan-CDK inhibitors like Palbociclib which blocks Rb phosphorylation is under pre-clinical phases and Ribociclib was terminated in phase I in conjunction with the analysis of cetuximab. More research is needed to decide whether the same form of CDK4/6 inhibitors are applicable.

COX-2 inhibitors

COX-2 catalyses the transformation of arachidonic acid into prostaglandins plays a role in epithelial tumour progression has been found to overexpress in oral premalignant lesions and oral cancer [87]. Celecoxib used as a cox-2 inhibitor and phase II is completed with inadequate dose performance. Celecoxib is currently active in phase II along with Erlotinib (NCT02748707). Phase II analysis of Celecoxib in oral premalignant lesions was completed, but the results have not yet been declared (NCT00014404).

mTOR inhibitors

Active PIK proceeds to the downstream effects, including AKT and mTOR activation, as identified in several tumours and known for tumour cell immortalization and increased proliferation. mTOR is activated in tumours and represents an attractive therapeutic target [115]. Phase II analysis of Rapamycin as an mTOR inhibitor with severe adverse effects like nephrotoxicity is completed. Temsirolimus, a Rapamycin analog is recruiting for phase I trial in combination with Bevacizumab (NCT01552434). Evorilimus is in phase I trial in combination with cetuximab to deal with OSCC (NCT01637194). Metformin a diabetes medicine is extensively studied in various trials to treat OSCC as it targets mTORC1 with a better survival rate in pre-clinical studies (NCT03685409, NCT02581137). BLY719, Copanlisib, BK120, Abemaciclib, Gedatolisib are some of the PI3K inhibitors studied in various trials (NCT02145312, NCT02822482, NCT03356587, NCT03065062).

EGFR inhibitors

EGFR overexpression is examined as a negative prognostic factor as it increases the tumor size, decreases radio sensitivity and increases the risk of recurrence. Two types of molecules differing slightly in the mode of action assist in the inhibition of EGFR targeted therapies. EGFR has been shown to be widely overexpressed along with poor prognosis [116]. Tyrosine kinase inhibitor Gefitinib is widely studied in combination with specific chemotherapeutics and radiotherapy in various trials. Phase I analysis of Gefitinib had been completed in combination with Paclitaxel, radiotherapy (NCT00083057). Phase II study, alone (NCT00024089) and in combination with Cisplatin, Flourouracil, Docetaxel (NCT00193284) has been completed, but its results are yet to be declared. Erlotinib is the class of EGFR inhibitor, which had undergone several phase I and II trials in combination with several drugs (NCT00101348, NCT01316757, NCT00055770, NCT00049283, NCT00030576). But the outcomes are unknown. Afatanib and Vandetanib are the other types of inhibitors that are under trials (NCT01783587, NCT00450138). In addition to Cetuximab, Panitumab and Zalutumab several regimens have gone through phase I, II trials (NCT00933387, NCT01445405, NCT00397384, NCT01264328, NCT00513383).

VEGF inhibitors

For a tumour to grow, invade and metastasize, angiogenesis is vital, which is the process of new blood vessel formation, so that tumour cells can acquire the required nutrients. Vascular endothelial cell growth factor (VEGF) has angiogenic effects in several cancers including OSCC [117]. Sorafenib, Dasatinib and Bevacizumab are the clinically investigated inhibitors of VEG-F in oral cancer. Bevacizumab and Lenalidomide are under phase I, in combination with Cetuximab (NCT01254617). Anlotinib is the other kind of VEGF inhibitor studied in-vitro in TCA8113 cells [118].

Immune checkpoint inhibitors

PD 1 is an inducible expressed immunoreceptor on T and B lymphocytes that acts as immune checkpoints while bound to PD-ligand1 (PD-L1). This results in the diminished T-cell activation leading to inhibit anti-tumour response and is overexpressed. Two FDA approved immunotherapies targeting immune checkpoints are currently available for the treatment of OSCC with platinum-based

Table 3 Clinical trial status and their outcomes

Targets/drugs	Trial number	Phase	Therapy	Status and outcome
CDK inhibitors				
Ribociclib	NCT02429089	I	Combination with Cetuximab in HPV (–) HNSCC	Myelosuppression, thrombocytopenia, and terminated early
COX-2 inhibitor				
Celecoxib	NCT02748707	II	Celecoxib, Erlotinib in OSCC	Active
Celecoxib	NCT00014404	II	Celecoxib in oral premalignant lesions	No results posted
Celecoxib	NCT02739204	II	Radiotherapy, Cisplatin, celecoxib in OSCC	Recruiting
mTOR inhibitors				
Rapamycin	NCT01195922	II	Sirolimus in cancer of oral cavity (II-IVA)	Thrombocytopenia
Temsirolimus	NCT01256385	II	Cetuximab, Temsirolimus in HNSCC including recurrent OSCC	6 cases were withdrawn due to toxicity. Progression free survival (PFS) is 105 days
Temsirolimus	NCT01552434	I	Bevacizumab, Temsirolimus	Recruiting
Everolimus	NCT01637194	I	Cetuximab, Everolimus in recurrent OSCC	No results posted
EGFR inhibitors				
Erlotinib	NCT00410826	II	Erlotinib, cisplatin, radiotherapy	Failed to increase PFS
Erlotinib	NCT00055913	I/II	Erlotinib, Bevacizumab	Diarrhoea and serious bleeding
Erlotinib	NCT00402779	III	Erlotinib	Cellulitis and failed to reduce the risk of oral cancer
Erlotinib	NCT00101348	I/II	Erlotinib, Cetuximab, Bevacizumab in recurrent oral cancer	No results posted
Erlotinib	NCT01316757	II	Erlotinib, Pacilitaxel, Cetuximab, Carboplatin	No results posted
Erlotinib	NCT00055770	I, II	Erlotinib, Docetaxel	No results posted
Erlotinib	NCT00049283	I	Erlotinib, Docetaxel, radiotherapy	No results posted
Erlotinib	NCT00030576	I, II	Cisplatin, Erlotinib	No results posted
Erlotinib	NCT00402779	III	Erlotinib for oral cancer	Results submitted
Lapatinib	NCT00424255	III	Lapatinib. Chemoradiation	Poor efficacy with nephrotoxicity
Lapatinib	NCT00114283	II	Lapatinib	No results posted
Gefitinib	NCT00068497	II	Gefitinib in HNSC including OSCC	No results posted
Gefitinib	NCT00083057	I	Gefitinib, Paclitaxel along with radiotherapy	No results posted
Gefitinib	NCT00352105	II	Cisplatin, Fluorouracil, Iressa, and Radiation Therapy	Blood lymphatic disorders, gastro-intestinal disorders
Gefitinib	NCT00193284	II	Gefitinib, Docetaxel, Cisplatin, Fluorouracil	No results posted
Gefitinib	NCT00024089	II	Gefitinib	No results posted
Afatinib	NCT01783587	I	Afatinib, Docetaxel, Radiotherapy	No results posted
Afatinib	NCT00514943	II	Afatinib, Cetuximab	No results posted
Vandetanib	NCT00450138	I	Vandetanib, Cisplatin, radiotherapy	No results posted
Cetuximab	NCT00933387	II	Cetuximab, Paclitaxel, Cisplatin	No results posted
Cetuximab	NCT01445405	I	Cetuximab, Cisplatin, radiotherapy	No results posted
Cetuximab	NCT00397384	I	Cetuximab, Erlotinib	No results posted
Panitumab	NCT00500760	II	Panitumab, Cisplatin and radiotherapy	No benefit and need reassessment
Panitumab	NCT01264328	II	Panitumab, Paclitaxel	No results posted
Panitumab	NCT00513383	I	Panitumab, Carboplatin, Paclitaxel, 5-Flourouracil, Cisplatin and Docetaxel	No results posted
Zalutumumab	NCT00382031	III	Zalutumumab	Do not increase overall survival
VEGF inhibitors				
Bevacizumab	NCT00409565	II	Bevacizumab, Cetuximab	Good treatment efficacy but less sample size
Dasatinib	NCT00507767	II	Dasatinib alone	PFS is 6.6%
Sorafenib	NCT00939627	II	Sorafenib, Cetuximab	Requires further validation
Sorafenib	NCT02035527	I	Sorafenib, Cisplatin, Docetaxel	No results posted
Lenalidomide	NCT01254617	I	Lenalidomide, Cetuximab	Active

Table 3 (continued)

Targets/drugs	Trial number	Phase	Therapy	Status and outcome
Immune checkpoint inhibitor				
Pembrolizumab	NCT02586207	I	Pembrolizumab, Cisplatin and radiation for squamous cell carcinoma of oral cavity	Active
Nivolumab	NCT03021993	II	Nivolumab alone for oral cavity SCC prior to surgery	Recruiting
Durvalumab	NCT03529422	I	Durvalumab, Tremelimumab, Radiotherapy	Recruiting
Durvalumab	NCT03174275	II	Durvalumab, carboplatin, nab-paclitaxel, cisplatin and RT	Recruiting
Durvalumab	NCT02827838	II	Durvalumab	Recruiting

chemotherapy: the programmed death inhibitors Nivolumab and Pembrolizumab [119]. NCT02586207, NCT03021993, NCT03529422, NCT03174275, NCT02827838 are the other trials that are recruited to evaluate the efficiency of Pembrolizumab, Nivolumab, Durvalumab in combination with other drugs.

Conclusion

Despite tremendous progress in the study of OSCC, there remain numerous gaps in our knowledge of molecular mechanisms. Based on the current research, it has been shown that tobacco and betelnut cause several genetic alterations to induce OSCC. Furthermore, the risk is increased by exposure to bacteria and viruses, whose mechanism of infection is complicated. Besides, the connection between inheritance and oral malignancy requires further investigation.

Multiple aberrations and signalling pathways are related to the action of oral cancer, but PI3K-AKT-GSK-3β axis is widely involved in many studies. Thus, drugs directly targeting AKT may be a possible option to treat OSCC. Several pre-clinical and clinical research targeting AKT are being studied. The mechanistic role of the majority of mutations such as Rb, Notch 1, circadian genes have poorly described. Besides the approval of many drugs, tumour resistance is a threatening phenomenon that hinders the success of treatment. As the drug resistance mechanism is complex, drugs targeting a single molecule is not enough and hence combinatorial drugs like Cetuximab and C-met inhibitor may result in newer opportunities in treatment.

Large clinical trials of various classes of drugs targeting several components of OSCC signalling pathways are being studied but phase III trials are lacking. On the other hand, few trials have failed to elicit a major response and require further investigation to overcome serious adverse effects. However, most of the findings are under investigation and the results of completed trials are awaited. Analyzing the genetic factors and their interactions would give details on its resistance mechanism and an idea about novel targets. This review focuses on the role of tumour suppressor genes, oncogenes, DNA repair genes, Immunomodulatory genes, Circadian genes, microRNA's and other genes accounting for cell proliferation, survival, angiogenesis and metastasis. The mechanism of the current treatment regimen and details of its resistance pattern are discussed in this review to provide a clear insight in developing new drugs and re-sensitize cancer cells to pre-existing drugs through combinatorial therapy. For future perspectives, upcoming clinical trials are summarized.

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Compliance with ethical standards

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